A DIRECT SUM DEVISSAGE PRESENTS THE MODULE AS THE SUM OF THE SUCCESSIVE QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the frozen snapshot stages the chapter's own preamble file beside it (retained-preamble.tex, 63 lines, 8116 bytes, the same shared Stacks-wide preamble the published chow and etale-cohomology cards use); the first line of the pinned chapter file is \input{preamble}, so no preamble text lives inside the chapter itself. The card takes that file's lines 8-18 (the verbatim package and the three \comment-based environments reference/slogan/history at lines 9-12, section 13.2, together with the Xy-pic setup) and its lines 154 to the runtime end of the file (the native theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- Property (3) of a direct sum devissage. Used at the start of the counted proof (lines 20634-20635). Supplies the inclusions of the successive quotients into the module.
- Transfinite induction over the well-ordered index set. Used in the middle of the counted proof. Shows the assembled map is injective by descending through the stages.

Lemma 0.1. *Let M be an R-module. If $(M_\alpha)_{\alpha\in S}$ is a direct sum dévissage of M, then $M \cong \bigoplus_{\alpha+1\in S} M_{\alpha+1}/M_\alpha$.*

Proof. By property (3) of a direct sum dévissage, there is an inclusion $M_{\alpha+1}/M_\alpha \to M$ for each $\alpha \in S$. Consider the map

$$f : \bigoplus\nolimits_{\alpha+1\in S} M_{\alpha+1}/M_\alpha \to M$$

given by the sum of these inclusions. Further consider the restrictions

$$f_\beta : \bigoplus\nolimits_{\alpha+1\leq\beta} M_{\alpha+1}/M_\alpha \longrightarrow M$$

for $\beta \in S$. Transfinite induction on S shows that the image of f_β is M_β. For $\beta = 0$ this is true by (0). If $\beta + 1$ is a successor ordinal and it is true for β, then it is true

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

for $\beta + 1$ by (3). And if β is a limit ordinal and it is true for $\alpha < \beta$, then it is true for β by (2). Hence f is surjective by (1).

Transfinite induction on S also shows that the restrictions f_β are injective. For $\beta = 0$ it is true. If $\beta+1$ is a successor ordinal and f_β is injective, then let x be in the kernel and write $x = (x_{\alpha+1})_{\alpha+1\leq\beta+1}$ in terms of its components $x_{\alpha+1} \in M_{\alpha+1}/M_\alpha$. By property (3) and the fact that the image of f_β is M_β both $(x_{\alpha+1})_{\alpha+1\leq\beta}$ and $x_{\beta+1}$ map to 0. Hence $x_{\beta+1} = 0$ and, by the assumption that the restriction f_β is injective also $x_{\alpha+1} = 0$ for every $\alpha + 1 \leq \beta$. So $x = 0$ and $f_{\beta+1}$ is injective. If β is a limit ordinal consider an element x of the kernel. Then x is already contained in the domain of f_α for some $\alpha < \beta$. Thus $x = 0$ which finishes the induction. We conclude that f is injective since f_β is for each $\beta \in S$. $\square$